CIRCLES AND REGULAR POLYGONS

11–1 INTRODUCTION

In this chapter we define what we mean by the *circumference* (or *length*) of a *circle* and then prove that the circumference C of a circle of radius r is given by the familiar formula, $C = 2\pi r$. This involves giving a definition of the number π. To do these things, we must learn a few facts about limits, because the study of circumference is made by approximating to the circle by polygons and considering polygons with a very large number of sides. It is therefore necessary, first, to prove some theorems about regular polygons.

11–2 CIRCLES AND REGULAR POLYGONS

Our study of the circle depends upon knowing about regular polygons inscribed in circles. We therefore study these matters first.

Definition 11–1. A regular polygon is *inscribed* in a circle if all the vertices of the polygon are on the circle. We also say that the circle *circumscribes the polygon*.

Exercise Group 11–1

1. With ruler and compass, inscribe a square in a given circle.
2. Using Exercise 1, show how to construct regular inscribed polygons of 8 sides; 16 sides; 32 sides.
3. With ruler and compass, inscribe a regular hexagon (6-sided polygon) in a given circle.
4. Using Exercise 3, show how to construct regular polygons of 3 sides; 12 sides; 24 sides; 48 sides.

Theorem 11–1. *There is exactly one circle circumscribing a given triangle.*

Proof. In Fig. 11–1, suppose that the triangle is $\triangle ABC$, and choose any two of the sides, say AB and BC. Let m and n be the perpendicular bisectors of these sides. The lines m and n must intersect at a point O, for if m and n did not intersect, they would be parallel, CB would be perpendicular to m, and the points A, B, and C would be collinear. The point O is equidistant from A and B, and also equidistant from B and C. (Why?) Hence $\overline{OA} = \overline{OB} = \overline{OC}$, and the circle with center O and radius $\overline{OA}$ passes through A, B, and C. This proves that there is at least one circle through A, B, and C. To see that there can be only one, we observe that the center of such a circle must be equidistant from A, B, and C, and hence must lie on the perpendicular bisectors of AB and BC. These lines, m and n, intersect in only one point. This completes the proof.

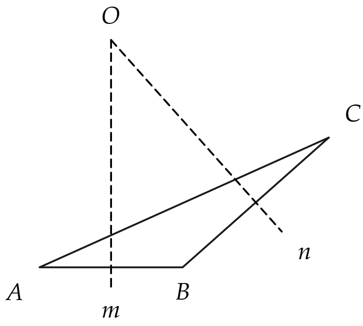

FIGURE 11–1

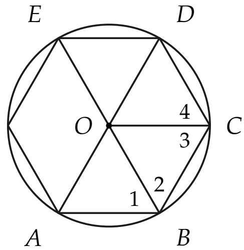

FIGURE 11–2

Theorem 11–2. *There is exactly one circle that circumscribes a regular polygon.*

Proof. In Fig. 11–2, suppose the polygon has successive vertices A, B, C, D, ... We show that the circle circumscribing $\triangle ABC$ actually circumscribes the polygon. Let O be the center of the circle circumscribing $\triangle ABC$, and consider the segments OA, OB, OC, OD, OE, ... By definition of circumscribing circle, $\overline{OA} = \overline{OB} = \overline{OC} = r$. We show that $\overline{OD} = r$. The same method can be used to prove that $\overline{OE} = \cdots = r$. Hence we need only show that $\overline{OD} = r$.

STATEMENTS	REASONS
1. $\angle 2 \cong \angle 3$.	Why?
2. $\angle ABC \cong \angle BCD$, and $AB \cong BC$.	The polygon is regular.
3. $\angle 1 \cong \angle 4$.	Addition and subtraction of angles postulate.
4. $\triangle OAB \cong \triangle ODC$.	S.A.S.
5. $OD \cong OA$.	Statement 4.
6. $\overline{OD} = r$.	Statement 5.

Exercise 11–2

Suppose that in Theorem 11–1 we had considered the perpendicular bisectors of AB and AC instead of AB and BC. Would the same point O have been found? What does the theorem say about the perpendicular bisectors of the sides of a triangle?

11–3 TANGENTS TO CIRCLES

To talk about circles *inscribed* in regular polygons, we need to know about tangents to circles.

Theorem 11–3. *If OP is a radius of a circle with center O, and a line l passes through P and is perpendicular to OP, then l has exactly one point in common with the circle.*

Proof. In Fig. 11–3, let A be any point on l different from P. Then $\triangle OPA$ is a right triangle, and $\overline{OA} > \overline{OP}$. (Why?) Hence A is outside the circle, and l intersects the circle only at the point P.

Definition 11–2. A line l is *tangent* to a circle if it has exactly one point in common with the circle; the point is called *the point of tangency*.

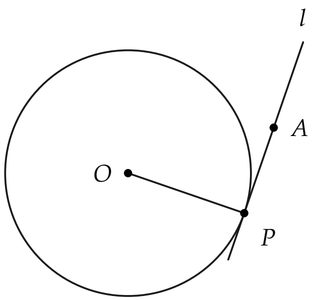

FIGURE 11–3

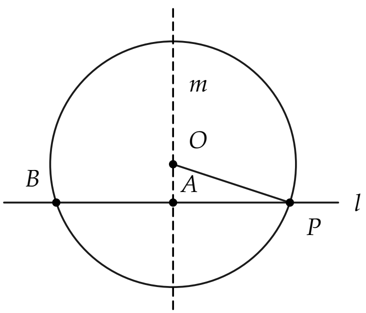

FIGURE 11–4

Theorem 11–4 Converse to Theorem 11–3. *A line tangent to a circle at a point P is perpendicular to the radius containing P.*

Proof. We will prove the contrapositive, that is, if the line is not perpendicular to the radius at P, then it is not tangent to the circle. Consider a circle (Fig. 11–4) with center O and a point P on the circle. Let l be a line through P, such that l is not perpendicular to OP.

STATEMENTS	REASONS
1. Let m be the line $\perp$ l passing through O. Let A be the intersection of m and l.	Why is there such a line?
2. Let B be the point on l on the opposite side of A from P, and such that $\overline{AB} = \overline{AP}$.	Why does such a point exist?
3. $\overline{OB} = \overline{OP}$.	Why? Several different arguments may be given.
4. B is on the circle and also on l.	Statements 2 and 3 and definition of a circle.
5. l intersects the circle twice, and hence l is not tangent to the circle. This completes the proof.	Statement 4 and definition of tangent.

Remark 1. In the above proof, the point A had to be inside the circle because $\overline{OA} < \overline{OP}$, by the Pythagorean theorem. The Pythagorean theorem shows that the perpendicular distance from a point to a line is less than the distance to any other point on the line.

Remark 2. An argument similar to the proof of Theorem 11–4, and using the Pythagorean theorem, can be made to show that a line can intersect a circle at most twice. Hence, a line will either be tangent to a circle, or intersect the circle in exactly two points, or fail to intersect the circle.

Exercise Group 11–3

1. Draw a circle, and construct (RC) the line tangent to the circle at a point of the circle.

*2. From a point outside a circle, construct a line tangent to the circle.

3. Prove that if A is a point not on a line l, and B and C are distinct points on l, with $AB \perp l$, then $\overline{AB} < \overline{AC}$.

Definition 11–3. A circle is said to be *inscribed* in a polygon, if it is tangent to each of the sides of the polygon. We also say that the polygon *circumscribes the circle*.

Theorem 11–5. *There is a circle inscribed in every regular polygon.*

STATEMENTS	REASONS
1. Let O be the center of the circle circumscribing the polygon (Fig. 11–5).	Theorem 11–2.
2. Let AB be a side of the polygon, and C the midpoint of AB.	AB has one, and only one, midpoint.
3. $OC \perp AB$.	Why?
4. The circle with center O and radius OC is tangent to AB at C.	Theorem 11–4.
5. Now let BD be a side of the polygon adjacent to AB, and E the midpoint of BD. Then $OE \cong OC$.	Why? This takes several steps.
6. Hence the circle with radius OC is also tangent to BD at E.	Statement 5 and the fact that $OE \perp BD$.

7. The above argument can be continued around the polygon to complete the proof.

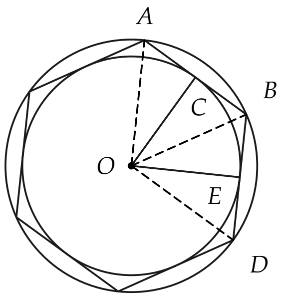

FIGURE 11–5

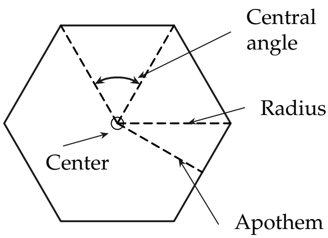

FIGURE 11–6

Exercise 11–4

Show that congruent chords in a circle are equidistant from the center of the circle.

Definition 11–4. The *center* of a regular polygon is the center of the inscribed or circumscribed circle. The *apothem* of a regular polygon is the length of the radius of the inscribed circle (that is, a number). The *radius* of a regular polygon is the radius of the circumscribed circle. A *central angle* of a regular polygon is an angle, with vertex at the center, whose sides contain adjacent vertices of the polygon. See Fig. 11–6.

Exercise Group 11–5

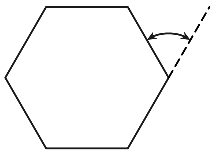

FIGURE 11–7

1. Inscribe a regular hexagon in a circle. Draw the circle inscribing the hexagon.
2. Using your experience in measuring angles in degrees (we will study this topic in the next chapter), how many degrees in a central angle of a regular pentagon? hexagon? n-gon?
3. What is the sum of the degree measures of the angles of a regular pentagon? hexagon? n-gon?
4. If an exterior angle of a regular polygon is as shown in Fig. 11–7, how many degrees in it if the polygon has 5 sides? 6 sides? n sides?
5. Prove that a central angle of a regular polygon is the

supplement of any angle of the polygon.

6. Show that an exterior angle of a regular polygon is congruent to a central angle.

7. In Definition 11–4, how do we know that the center of the inscribed polygon and the center of the circumscribed polygon are identical?

11–4 REGULAR POLYGONS AND SIMILARITY

All our proofs about area and circumference of circles depend upon the following theorem.

Theorem 11–6. *Two regular polygons are similar if, and only if, they have the same number of sides.*

Preliminary remark. Recall that two polygons are similar if there is a correspondence between their angles and sides, such that corresponding angles are congruent and corresponding sides are proportional. Thus, if the polygons are similar, they necessarily have the same number of sides, and the "only if" part of the theorem is obvious. Therefore, we have only to prove that if they have the same number of sides, they must be similar.

Proof. Suppose that two regular polygons have consecutive vertices $A, B, C, D, E, \ldots$ and $A', B', C', D', E', \ldots$, respectively, as in Fig. 11–8. Suppose also that the polygons are not similar. Since they are regular,

$$\frac{\overline{AB}}{\overline{A'B'}} = \frac{\overline{BC}}{\overline{B'C'}} = \frac{\overline{CD}}{\overline{C'D'}} = \frac{\overline{DE}}{\overline{D'E'}} = \cdots .$$

It must be, then, that $\angle A \ncong \angle A'$, $\angle B \ncong \angle B'$, etc. If $\angle A \ncong \angle A'$, then $\angle A > \angle A'$ or $\angle A < \angle A'$. Suppose that $\angle A < \angle A'$. Since the polygons are regular, $\angle B < \angle B'$, $\angle C < \angle C'$, etc. The exterior angle at A is greater than the exterior angle at A'. (Why?) This is true also for the exterior angles at B and B', C and C', etc. The sum of n exterior angles in polygon $ABCDE\ldots$ is two straight angles. Clearly then, the sum of n exterior angles in polygon $A'B'C'D'E'\ldots$ is not two straight angles. This contradiction completes the proof.

Remark. If we had studied measurement of angles, we could more easily have proved that $\angle AOB \cong \angle A'O'B'$, because the measure of these angles depends only upon the number of sides n. In degrees, the measure of $\angle AOB$ would be equal to $360/n$.

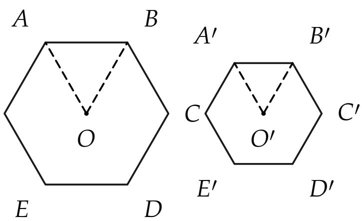

FIGURE 11–8

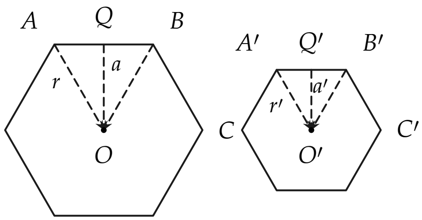

FIGURE 11–9

Theorem 11–7. *The perimeters of regular polygons, with the same number of sides, are proportional to their radii (or apothems).*

Proof. Suppose that two polygons (of n sides) are labeled as in Fig. 11–9, and have perimeters p and p'. We must show that $p/p' = r/r' = a/a'$.

STATEMENTS	REASONS
1. $\triangle AOQ \sim \triangle A'O'Q'$.	Why?
2. $r/r' = a/a' = \overline{AQ}/\overline{A'Q'}$.	Statement 1.
3. $p' = n \cdot \overline{A'B'} = 2n \cdot \overline{A'Q'}$, and $p = n \cdot \overline{AB} = 2n \cdot \overline{AQ}$.	Definition of perimeter and the fact that Q and Q' are midpoints.
4. $p/p' = \overline{AQ}/\overline{A'Q'}$.	Statement 3 and algebra.
5. $p/p' = r/r' = a/a'$.	Statements 2 and 4.

Theorem 11–8. *The area of a regular polygon is equal to one-half the product of apothem and perimeter; area* $= \frac{1}{2}a \cdot p$.

The proof is left as an exercise.

Theorem 11–9. *The areas of two regular polygons, with the same number of sides, are proportional to the squares of their radii (or the squares of their apothems, or the squares of their sides).*

The proof is left as an exercise.

Exercise Group 11–6

1. Two regular heptagons (7-gons) have sides 3 in. and 4 in., respectively. What is the ratio of their perimeters? of their areas?

2. A regular pentagon has twice the area of another regular pentagon. What is the ratio of their perimeters?

3. Find the radius and apothem of an equilateral triangle of side 2 in.

4. Two regular polygons of the same number of sides have perimeters of 28 and 42 in. The radius of the first polygon is x in. What is the radius of the second?

5. The apothems of two regu-

lar hexagons are 3 in. and 5 in. What is the ratio of their areas?

6. The area of a regular 17-gon is 16/9 times the area of a second 17-gon. What is the ratio of their perimeters? radii?

7. If a regular hexagon and an equilateral triangle are inscribed in a circle, the side of the hexagon is twice the apothem of the triangle. Show this.

8. Find the areas of the squares inscribed in and circumscribed about a circle of radius 1. Also find the perimeters of the squares.

9. Repeat Exercise 8, using hexagons instead of squares.

11–5 LIMITS

To investigate circumference and area of circles, it is necessary for us to discuss *limits*. Precise definitions relating to limits are difficult to grasp, and a careful study is first made in college calculus courses. For our purposes, only a little information about limits is required. All our statements about limits are statements about *numbers*. Therefore you may regard these statements as giving information about numbers that we will *assume* or postulate.

Definition 11–5. Suppose that for each positive integer n, there is associated in some way, a number x_n. The numbers $x_1, x_2, \ldots, x_n, \ldots$ then form an infinite sequence. The number x_n is called the nth term of the sequence. We shall indicate the sequence by $\{x_n\}$.

For example, let the nth term of a sequence be given by $x_n = 2n$. The positive integer 1 gives the first term $2 \cdot 1$. The positive integer 2 gives the second term $2 \cdot 2$. The third term is $2 \cdot 3$, and so on. The sequence given by $x_n = 2n$, then, would look like this:

$$\begin{array}{rcccccccc}
\text{Positive integers} \rightarrow & 1 & 2 & 3 & 4 & \dots & n & \dots \\
& \updownarrow & \updownarrow & \updownarrow & \updownarrow & & \updownarrow & \\
\text{Terms of the sequence} \rightarrow & 2 & 4 & 6 & 8 & \dots & 2n & \dots.
\end{array}$$

Exercise Group 11–7

1. Write the first six terms of the sequence whose nth term is given by $x_n = 1/n$; by $x_n = \frac{1}{2}n$; by $x_n = \frac{1}{5}n$.
2. Write the 100th term for each of the sequences in Exercise 1. When n is very large, what number are the terms close to?
3. Write the first 10 terms of the sequence whose nth term is $x_n = (n+1)/n$. Write the 100th term; the 1000th. What number are these terms close to when n is extremely large?
4. A sequence is given by $x_n = 2/(n+3)$. Write the first six terms; the 100th term; the millionth term. Is there a number that the terms of the sequence are close to for large values of n?
5. Consider a regular polygon of n sides inscribed in a circle. Let p_n be the perimeter of the polygon. Does this define a sequence?
6. A sequence is given by $x_n = 2/\sqrt{n}$. Write the first six terms. Is there a number that the terms of the sequence are close to for large values of n?
7. Imagine $\sqrt{2}$ written as an infinite decimal, $\sqrt{2} = 1.4142\ldots$. Define a sequence x_n as follows: $x_n =$ the decimal obtained by cutting off the decimal expansion of $\sqrt{2}$ after the nth decimal place. Then $x_1 = 1.4$, $x_2 = 1.41$, etc. What is x_4? x_5? To what number is x_n close when n is large?

We can now define the *limit* of a sequence, an idea that should be clear from the above exercises. We shall be content with expressing this idea in an intuitive way, but it should be mentioned that a more complicated definition is necessary in order to *prove* theorems about limits. We *assume* that the theorems we need concerning limits are true.

Definition 11–6. The number L is called the *limit* of the sequence $\{x_n\}$ if, for large values of n, the term x_n is necessarily arbitrarily close to L. We then say that L is *the limit of the sequence of terms* x_n and write "$L = \lim \{x_n\}$."

Exercise Group 11–8

In each of the following problems, the nth term of a sequence is given. You are to decide by inspection what the limit of the sequence is. Note that a sequence does not have to possess a limit.

1. $x_n = 1/n$.
2. $x_n = 1/10n$.
3. $x_n = (n+1)/n$.
4. $x_n = \sqrt{n}$.
5. $x_n = 5$.
6. $x_n = 2n/(n+1)$.
7. $x_n = n+2$.

*8. $x_n = (2n+100)/(4n+50)$.

*9. $x_n = (n+1)/n^2$.

*10. $x_n = (n^2+5n+6)/[2n(n+2)]$.

When we discuss the circumference and area of circles, we will need the following two theorems on limits. We assume that these theorems are true, and thus treat them as postulates. But they are not postulates about our *geometry*; they are postulates about the *number system*.

Theorem 11–10. *Consider two sequences with nth terms x_n and y_n, respectively, and suppose that* $\lim \{x_n\} = X$ *and* $\lim \{y_n\} = Y$. *Then* *the sequence whose nth term is $x_n y_n$ has a limit, and* $\lim \{x_n y_n\} = XY$.

Theorem 11–11. *With the same notation as in Theorem 11–10, and the additional hypothesis that y_n and Y are not zero, then the sequence whose nth term is x_n/y_n has a limit, and* $\lim \{x_n/y_n\} = X/Y$.

Exercise Group 11–9

1. Verify Theorem 11–10 if $x_n = (2n+1)/n$ and $y_n = n/(n+1)$.
2. If $\lim \{x_n\} = 2$, what is $\lim \{(x_n)^2\}$?
3. If $\lim \{x_n\} = 3$ and $\lim \{y_n\} = 4$, what is $\lim \{x_n/y_n\}$? $\lim \{x_n y_n\}$? $\lim \{y_n/x_n\}$?
4. If x_n is "very close" to 5 and y_n is "very close" to 7, what is $x_n y_n$ close to? x_n/y_n? y_n/x_n?

11–6 CIRCUMFERENCE

From your everyday experience, you already have an idea of the perimeter or circumference of a circle. For example, if you were asked to find the circumference of a wheel, you might wrap a string around the wheel, then pull it out straight, and measure its length. But such a practical method is not adequate for geometry, since we cannot describe this process in our geometry. It is necessary that we first *define* circumference. The definition must satisfy our intuition about what we *feel* the circumference should be, and should be one that can readily be *used* to prove theorems.

Observe that we had no difficulty in defining the perimeter of a polygon, because the polygon consisted of a finite number of segments, each of which had a length. A circle contains no segment of a line, so we cannot define its perimeter so simply.

To arrive at our definition, we need the following theorem, the truth of which we assume.

Theorem 11–12. *For a given circle, let p_n be the perimeter of a regular polygon of n sides inscribed in the circle. The numbers p_n $(n = 3, 4, 5, \ldots)$ form a sequence that has a limit.*

Remarks. The proof of this theorem involves two steps. First, it is shown that the sequence is increasing, that is, each term is less than the one that follows it. Then it is shown that the sequence is *bounded*, that is, the terms do not become indefinitely large. It then follows, as a fundamental property of real numbers, that such a sequence has a limit.

It is natural to ask how we know that regular polygons with n sides exist for all positive integers $n \geqq 3$. We do know that there are regular polygons with the number of sides equal to 2^2, 2^3, 2^4, etc., for we can inscribe a square in the circle, and, by bisecting the central angles, we get a regular octagon, and then a regular 16-gon, and so on, as in Fig. 11–10.

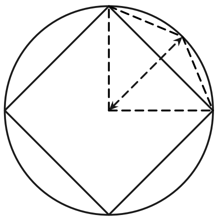

FIGURE 11–10

Therefore, the perimeters of these special regular polygons form a sequence $p_4, p_8, p_{16}, p_{32}, \ldots$, and by Theorem 11–12 and the definition of limit, this sequence has a limit.

We can now give a definition of circumference. Observe that in order to *define* the circumference of a circle, we are *necessarily forced* to use limits.

Definition 11–7. The *circumference* C of a circle is the limit of the sequence of perimeters p_n of regular inscribed polygons. $C = \lim\{p_n\}$.

Exercise Group 11–10

1. For a given circle, show that the perimeter of an inscribed square is less than that of an inscribed octagon.
2. Find the perimeter of a square inscribed in a circle of radius 1. Hence show that the circumference of the circle is more than $4\sqrt{2}$.
3. By inscribing a regular hexagon in a circle of radius 1, show that the circumference of the circle is more than 6.

The fundamental result on the circumferences of circles is the following:

Theorem 11–13. *The circumferences of two circles are proportional to their radii (or to their diameters). That is, if the circles have circumferences C and C', radii r and r', and diameters d and d', then $C/C' = r/r' = d/d'$.*

Proof.

STATEMENTS	REASONS
1. $C/C' = \lim \{p_n\} / \lim \{p'_n\}$, where p_n, p'_n, are the perimeters of the regular inscribed polygons of n sides in the circles.	Definition 11–7.
2. $C/C' = \lim \{p_n/p'_n\}$.	Theorem 11–11 on limits.
3. $p_n/p'_n = r/r' = d/d'$.	Theorem 11–7.
4. $\lim \{p_n/p'_n\} = \lim \{r/r'\}$.	$p_n/p'_n = r/r'$ for all n.
5. $\lim \{r/r'\} = r/r'$.	r and r' are the same for all n.
6. $\lim \{p_n/p'_n\} = r/r' = d/d'$.	Statements 3, 4, and 5.
7. $C/C' = r/r' = d/d'$.	Statements 2 and 6.

Remark. Observe that this theorem about circumferences is basically a theorem about similar polygons.

Theorem 11–14. *If C is the circumference of a circle, and d is its diameter, the ratio C/d is the same for all circles.*

Proof. This theorem is really just a corollary to Theorem 11–13. But it is so important that it is given a place by itself. The proof is as follows. From Theorem 11–13, we have $C/C' = d/d'$. This proportion implies the proportion $C/d = C'/d'$. This last equality says that C/d is the same number for any two circles. This completes the proof.

Definition 11–8. The number C/d, which is the same for all circles, will be denoted by the Greek letter π (pi).

Corollary 11–14–1. $C = \pi d = 2\pi r$.

Remark. The definition of π does not immediately give its numerical value. So a problem remains—namely that of calculating

π to any desired degree of accuracy. We examine later a bit of the history of this number and methods for calculating its value. At the moment, and for purposes of the exercises, we mention two approximate values commonly used for π, namely 22/7 and 3.1416.

Exercise Group 11–11

1. Find, approximately, the circumference of a circle whose radius is 7 in.; 2.131 in.; 49 ft.
2. Find, approximately, the radius of a circle whose circumference is 66 ft; 12.125 in.; 100.00 ft.
3. Give an exact expression for the area of a square whose perimeter is equal to the circumference of a circle of radius 1.
4. Suppose that a circular steel band surrounds the earth at the equator (radius = 4000 mi). If the band is made longer by inserting a piece 1 ft long, about how far will the band be from its previous position?
5. A tire on a car has a diameter of 26 in. If it revolves at 10 revolutions per second, what is the approximate speed of the car in miles per hour?
6. About how many revolutions per minute does a car wheel make (diameter = 28 in.) when the car is traveling at 60 mph?
7. The radius of one circle is three times the radius of a second. How do their circumferences compare?

11–7 COMPUTATION OF PI

There are many ways of computing π, some much more convenient than others. The obvious way is to compute the perimeters

p_n of regular polygons with a large number of sides. Then π would be given approximately by p_n/d. This is a rather tedious process, and we use it here to get a very rough estimate of π.

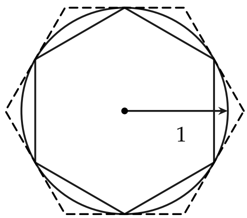

FIGURE 11–11

Consider a regular inscribed hexagon in a circle of radius 1, as shown in Fig. 11–11. The perimeter of the hexagon is 6. Increasing the number of sides will increase the perimeter. Hence $C > 6$, so that $\pi = C/d > 6/2 = 3$, and $\pi > 3$.

If we consider a regular circumscribed hexagon, we find that its perimeter is $4\sqrt{3}$, and since C is less than this perimeter,

$$\pi = \frac{C}{d} < \frac{4\sqrt{3}}{2} = 2\sqrt{3} < 3.465,$$

and $\pi < 3.465$.

In this way, we have obtained a very rough value for π. It lies somewhere between 3 and 3.5. By these same methods, Archimedes (3rd century B.C.) showed that π is between 3 1/7 and 3 10/71. The Babylonians used 3 as the value of π, and there is a verse in the Bible that implies that the writer believed $\pi = 3$.

By far the most convenient and rapid methods for calculating π are based on formulas derived in calculus.* We cannot take

*A formula from calculus that can be used to calculate π is

$$\pi = 4\left(1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \frac{1}{9} - \frac{1}{11} + \cdots\right).$$

up these formulas here, but it is interesting to know that in the 17th century a Scot by the name of Shanks,† using one of these formulas, computed π to 707 decimal places. His reasons for doing this are at best obscure. More recently, a group working on one of the large electronic computers set up a program for the calculation of π on the computer and obtained, in a few hours, π to more than 2000 decimal places! This demonstrates the extreme speed of modern computing machines. The corresponding job by hand would take hundreds of years. The correct value of π to 14 decimal places is 3.14159265358979.

The number π is an irrational number. A proof of this is not easy, and was given first by the German mathematician Johann Lambert in 1766.

Exercise Group 11–12

1. By considering inscribed and circumscribed squares, show that $2\sqrt{2} < \pi < 4$.

*2. By considering an inscribed regular octagon, show that $\pi > 3.06$.

*3. Find the perimeter of a regular 12-gon in a circle of radius 1. Hence find an approximate value for π. Can you continue to find the perimeter of a regular 24-gon?

4. What does the irrationality of π imply about its decimal representation?

11–8 AREA OF CIRCLES

In studying area we first obtained the area of a rectangle by counting squares. Areas of triangles may be obtained by forming parallelograms and observing that the triangle is half the area of the

†One of your authors makes no claim to be a descendant of this misguided computer.

parallelograms. Areas of polygons were obtained by cutting up the polygons into triangles. To define what we mean by the area of a circle, it is necessary to use again the idea of limit, just as it was necessary to use limits in order to define circumference. We all have a fairly clear idea of what we *feel* the area of a circle should be, and we use this impression to define area. Since the area of the circle should be close to the area of the inscribed polygon with a large number of sides, we first need to know that the limit of the areas of inscribed polygons exists. This fact is explicitly stated as a theorem below, which we accept without proof.

Theorem 11–15. *Given a circle C, let S_n denote the area of an inscribed regular polygon of n sides; then* $\lim\{S_n\}$ *exists. In other words, there is a number S, such that the term S_n is arbitrarily close to S for large n.*

Remark 1. We know, from Section 11–6, that regular polygons exist with the number of sides equal to powers of 2. In using Theorem 11–15, we can consider only polygons that we know to exist.

Remark 2. Theorem 11–15 is a theorem about numbers. It has little geometric content, so we need not feel that it is another geometric postulate. If we consider only polygons with 4, 8, 16, 32,... sides, it is easy to show that each area in our sequence S_4, S_8, S_{16}, $S_{32}, \ldots$ is less than the one that follows. On the other hand, we can see that each of these polygons has an area less than the area of a circumscribed square. This indicates that our sequence is not only increasing, but is "bounded above." It is a theorem about numbers that such a sequence has a limit.

Definition 11–9. The area S of a circle is the limit of the areas of inscribed regular polygons. That is, $S = \lim\{S_n\}$.

We can now prove the main theorem about the area of a circle. But because the definition of area uses the idea of limit, we must use some theorems about limits. In particular, we need to know the following fact about the apothems of the inscribed polygons.

Theorem 11–16. *Let a_n be the apothem of an inscribed regular polygon of n sides in a circle of radius r, then* $\lim \{a_n\} = r$.

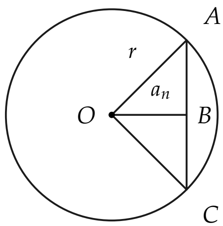

FIGURE 11–12

Proof. In Fig. 11–12, $\overline{AC}$ is the length of a side of the polygon, and is a small number when the number of sides of the polygon is large. By the Pythagorean theorem, $a_n = \sqrt{r^2 - \overline{AB}^2}$. Thus, when n is very large, $\overline{AB}^2$ is very small, and $r^2 - \overline{AB}^2$ is very near r^2. Hence a_n will be near r. This is exactly what we mean when we say that $\lim \{a_n\} = r$.

Theorem 11–17. *The area of a circle is equal to one-half the circumference times the radius.* $S = \frac{1}{2}$ *(circumference)* $\times$ *(radius).*

Proof.

STATEMENTS	REASONS
1. $S = \lim\{S_n\}$, where S_n is the area of the regular inscribed polygon of n sides.	Definition.
2. $S_n = \frac{1}{2}p_n \cdot a_n$, where p_n is the perimeter, and a_n is the apothem, of the polygon.	Theorem 11–8.
3. $\lim\{S_n\} = \frac{1}{2}(\lim\{p_n\}) \cdot (\lim\{a_n\})$.	Statement 2 and Theorem 11–10 on limits.
4. $\lim\{p_n\} = C =$ circumference.	Definition of circumference.
5. $\lim\{a_n\} = r =$ radius.	Theorem 11–16.
6. $\lim\{S_n\} = \frac{1}{2}C \cdot r$.	Statements 3, 4, and 5.
7. $S = \frac{1}{2}C \cdot r$.	Statements 1 and 6.

Corollary 11–17–1. *The area of a circle of radius r is πr^2.*

Corollary 11–17–2. *The areas of two circles are proportional to the squares of their radii (or diameters).*

The proofs are left to the student.

Exercise Group 11–13

1. Two circles have radii of 5 in. and 2.5 in. Find their areas. What is the ratio of their areas?

2. An annular ring (Fig. 11–13) has inner radius 2 ft and outer radius 4 ft. Find its area.

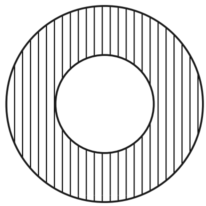

FIGURE 11–13

3. Figure 11–14 is composed of a circle and semicircles. If $\overline{AB} = \overline{BC} = 2$, B is the center of the circle, and AC is a diameter, find the area of the shaded portion.

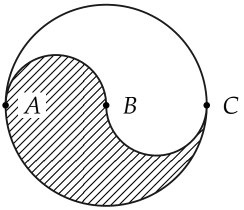

FIGURE 11–14

4. Show that the area of the inscribed circle is half the area of the circumscribed circle of the square (Fig. 11–15).

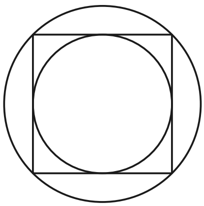

FIGURE 11–15

5. The area of a circle is 154 in^2. Find its radius. (Use $\pi = 22/7$.)

6. The area of a circle is 100 ft^2. Find its radius.

7. Semicircles are constructed on the sides of a right triangle, as in Fig. 11–16. Show that the area of the largest is the sum of the areas of the other two.

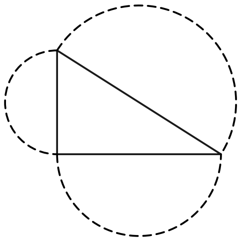

FIGURE 11–16

8. A circle has its circumference equal to its area. What is its radius?

9. A rectangular barn is 30 ft by 40 ft. A goat is tethered at one corner by a rope 40 ft long. Over what area can he graze? Over what area can he graze if the rope is 70 ft long?

10. The circumference of a circle is 50 ft. What is its area?

11. The area of a circle is 288 in^2. What is the area if the unit of length is a foot?

12. On Mars the standard unit of length is the *pfutth*, which is 0.88 as long as our foot. A Martian states that the area of a circle is 16 square *pfutths*. What is the area in square feet?

*13. Show that the area of a circle is the limit of the areas of circumscribed regular polygons.

14. The area of a circle is 49/4 times the area of another circle. What is the ratio of their radii?

REVIEW OF CHAPTER 11

The following definitions were given in this chapter. Be sure that you know them.

inscribed polygon (circle)
circumscribed polygon (circle)
tangent
center, radius, apothem of regular polygons
sequence, limit
circumference
π
area of a circle

The following is a list of the more important theorems of the chapter. Give complete statements of these theorems. Then try to give a proof for each. Do not refer to the text until you need help.

1. A regular polygon has a circumscribed circle.

2. A line tangent to a circle is perpendicular to a radius.

3. Converse of 2 (almost).

4. Regular polygons are similar if and only if they have the same number of sides.
5. The perimeters of similar polygons are proportional to their radii.
6. The area of a regular polygon is one half the product of apothem and perimeter.
7. The circumferences of two circles are proportional to their radii.
8. The area of a circle is equal to one half the circumference times the radius.

Miscellaneous Questions

1. Why is it important in this chapter to know that regular polygons of the same number of sides are similar?
2. Why is it important that circumference is proportional to the radius? (Tell why this enables us to define π.) This theorem depends on what theorems on similar polygons?
3. Tell why we must *define* circumference and area for circles.
4. In proving that area $= \pi r^2$, what theorems about polygons did we need? what theorems on limits?